15. option(b)

$\text{rate}_1 = k[0.1]^n[0.1]^m$ ———(1)

$\text{rate}_2 = k[0.2]^n[0.1]^m$ ———(2)

$(2)/(1)$

$$\frac{2x}{x} = \frac{k[0.2]^n[0.1]^m}{k[0.1]^n[0.1]^m}$$

$$\frac{2x}{x} = 2^n \quad \therefore n = 1$$

$\text{rate}_3 = k[0.1]^n[0.2]^m$ ———(3)

$\text{rate}_4 = k[0.2]^n[0.2]^m$ ———(4)

$(4)/(2)$

$$\frac{8x}{2x} = \frac{k[0.2]^n[0.2]^m}{k[0.2]^n[0.1]^m}$$

$$\frac{8}{2} = 2^m \quad \therefore m = 2$$

$\therefore$ rate $= k[A]^1[B]^2$

16. option(c)

For a first reaction, when the concentration of reactant is doubled, then the rate of reaction also doubled.

Rate constant is independent of concentration and is a constant at a constant temperature, i.e., it depends on the temperature and hence, it will not be doubled and when the concentration of the reactant is doubled.

17. option(a)

The unit of rate constant is s^{-1} and it indicates that the reaction is first order.

18. option(c)

$$Rate = \frac{d[N_2O_5]}{dt} = \left(\frac{1}{2}\right)\frac{d[NO_2]}{dt} = \frac{2\,d[O_2]}{dt}$$

Given that

$$\frac{d[N_2O_5]}{dt} = 6.5 \times 10^{-2} \text{ mol } L^{-1}s^{-1}$$

$$\frac{d[NO_2]}{dt} = 2 \times 6.5 \times 10^{-2} = 1.3 \times 10^{-1} \text{ mol } L^{-1}s^{-1}$$

$$\frac{d[O_2]}{dt} = \frac{6.5 \times 10^{-2}}{2} = 3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}$$

19. option(d)

$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$

$$Rate = \frac{-d[H_2O_2]}{dt} = \frac{d[H_2O]}{dt} = \frac{2d[O_2]}{dt}$$

no of moles of oxygen $= \left(\frac{48}{32}\right) = 1.5$ mol

$\therefore$ rate of formation of oxygen = 2 1.5 = 3 mol min^{-1}

20. option(a)

For a first order reaction $t_{1/2}$ is independent of initial concentration .i.e., $\therefore n \neq 1$; for such cases

$$t_{1/2} \alpha \frac{1}{[A_0]^{n-1}} \quad \text{------- (1)}$$

If $[A_0] = 2[A_0]$; then $t_{1/2} = 2t_{1/2}$

$$2\,t_{1/2} \alpha \frac{1}{[2\,A_0]^{n-1}} \quad \text{------ (2)}$$

$$\frac{(2)}{(1)} \Rightarrow$$

$$2 = \frac{1}{[2A_0]^{n-1}} \times \frac{[A_0]^{n-1}}{1}$$

$$2 = \frac{[A_0]^{n-1}}{[2A_0]^{n-1}}$$

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